

Loop

Loop is used when you want to run same block of code multiple times without writing same block of code again and again

```
In [1]: ▶ print("data science")
          print("data science")
          print("data science")
          print("data science")
          print("data science")
```

```
data science
data science
data science
data science
data science
```

```
In [2]: ▶ i=1      # initializing
          while i<=5:
              print('data science',i)
              i=i+1    #increment
```

```
data science 1
data science 2
data science 3
data science 4
data science 5
```

```
In [3]: ▶ i=5
          while i>=1:
              print("data science",i)
              i=i-1    #decrement
```

```
data science 5
data science 4
data science 3
data science 2
data science 1
```

Nested while loop

nested while loop means while loop inside the another while loop this means first it run the inside loop completely for every single iteration of outer loop

```
In [4]: ▶ i=1
while i<=5:
    print('data science')
    j=1
    while j<=4:
        print('technology')
        j=j+1
    i=i+1
    print()
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
data science
technology
technology
technology
technology
```

```
In [5]: ▶ i=1
while i<=5:
    print('data science',end=" ")
    j=1
    while j<=4:
        print('technology',end=" ")
        j=j+1
    i=i+1
    print()
```

```
data science technology technology technology technology
data science technology technology technology technology
data science technology technology technology technology
data science technology technology technology technology
data science technology technology technology technology
```

```
In [6]: ▶ i=1
while i<=2:
    j=0
    while j<=2:
        print(i*j,end=" ")
        j+=1
    print()
    i+=1
```

```
0 1 2
0 2 4
```

```
In [7]: ▶ i=1
while i<=4:
    j=0
    while j<=3:
        print(i*j,end=" ")
        j+=1
    i+=1
    print()
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

FOR LOOP -

normally while loop it work with iteration or certaion some condition but for loop it will work with sequence (list, string,int)

for loop that allows you to run block of code in specific number of times

```
In [8]: ▶ n=4
        for i in range(0,n):
            print(i)
```

```
0
1
2
3
```

iterating over list,tuple,string and dictionary using for loop in python

```
In [9]: ▶ li=["akshitha",'perumandla',19]
        li.append(20)
        for i in li:
            print(li)
```

```
['akshitha', 'perumandla', 19, 20]
['akshitha', 'perumandla', 19, 20]
['akshitha', 'perumandla', 19, 20]
['akshitha', 'perumandla', 19, 20]
```

```
In [10]: ▶ tup=("akshitha","perumandla",19)

        for i in tup:
            print(tup)
```

```
('akshitha', 'perumandla', 19)
('akshitha', 'perumandla', 19)
('akshitha', 'perumandla', 19)
```

```
In [11]: ▶ s="Geeks"
        for i in s:
            print(i,end=" ")
```

```
G e e k s
```

```
In [12]: ▶ d=dict({'x':123, 'y':354})
        for i in d:
            print("%s %d"%(i,d[i]))
```

```
x 123
y 354
```

```
In [13]: ➤ set1={1,7,3,4,5,6}
          set1.add(10)
          for i in set1:
              print(i)
```

```
1
3
4
5
6
7
10
```

```
In [14]: ➤ li=["geeks","for","geeks"]
          for index in range(len(li)):
              print(li[index])
```

```
geeks
for
geeks
```

The code iterates over a sequence of list and prints each element after loop ends it print else block when loop completes without break

```
In [15]: ➤ li=["geeks","for","geeks"]
          for i in range(len(li)):
              print(li[i])
          else:
              print("inside else block")
```

```
geeks
for
geeks
inside else block
```

```
In [16]: ➤ li=["geeks","for","geeks"]
          for i in range(3):
              print(li[i])
          else:
              print("inside else block")
```

```
geeks
for
geeks
inside else block
```

```
In [17]: ▶ li=["geeks","for","geeks"]
        for i in range(3):
            print(li[i])
            break
        else:
            print("inside else block")
```

geeks

```
In [18]: ▶ name1=[1,3.5,'hello']
        for i in name1:
            print(i)
```

1
3.5
hello

```
In [19]: ▶ for i in [2,3,7.8,'hi']:
        print(i)
```

2
3
7.8
hi

```
In [20]: ▶ range(5)
```

Out[20]: range(0, 5)

```
In [21]: ▶ for i in range(5):
        print(i)
```

0
1
2
3
4

```
In [22]: ▶ for i in range(2,5):
        print(i)
```

2
3
4

```
In [23]: ▶ for i in range(1,10,3):
        print(i)
```

1
4
7

```
In [24]: ▶ for i in range(1,21):  
          print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20
```

```
In [25]: ▶ for i in range(1,51):  
          if i%5==0:  
              print(i)
```

```
5  
10  
15  
20  
25  
30  
35  
40  
45  
50
```

```
In [26]: ▶ fruits=["apple","orange","kiwi"]  
          for fruit in fruits:  
              print(fruit)
```

```
apple  
orange  
kiwi
```

```
In [27]: ▶ num=5  
for i in range(1,11):  
    print('5','*',i,'=',num*i)
```

```
5 * 1 = 5  
5 * 2 = 10  
5 * 3 = 15  
5 * 4 = 20  
5 * 5 = 25  
5 * 6 = 30  
5 * 7 = 35  
5 * 8 = 40  
5 * 9 = 45  
5 * 10 = 50
```



```
In [28]: ► for i in range(1,51):  
           if i%5!=0:  
               print(i)
```

```
1  
2  
3  
4  
6  
7  
8  
9  
11  
12  
13  
14  
16  
17  
18  
19  
21  
22  
23  
24  
26  
27  
28  
29  
31  
32  
33  
34  
36  
37  
38  
39  
41  
42  
43  
44  
46  
47  
48  
49
```

while loop is used to execute a block of statement repeatedly until the condition is satisfied

```
In [29]: ▶ x=int(input('How many chocolates you want: ?'))
          i=1
          while i<=x:
              print('chocolate')
              i+=1
```

How many chocolates you want: ?5
chocolate
chocolate
chocolate
chocolate
chocolate

```
In [30]: ▶ available=5
          x=int(input('How many chocolates you want: ?'))
          i=1
          while i<=x:
              print('chocolate')
              i+=1
```

How many chocolates you want: ?5
chocolate
chocolate
chocolate
chocolate
chocolate

```
In [31]: ▶ cnt=0
          while(cnt<3):
              cnt=cnt+1
              print("Hello Akshitha")
```

Hello Akshitha
Hello Akshitha
Hello Akshitha

```
In [32]: ▶ cnt=0
          while(cnt<3):
              print("Hello Akshitha")
              cnt=cnt+1
```

Hello Akshitha
Hello Akshitha
Hello Akshitha

using else statement with while loop in python

```
In [33]: ▶ cnt=0
while(cnt<3):
    cnt=cnt+1
    print("Hello Akshitha")
else:
    print("in else block")
```

```
Hello Akshitha
Hello Akshitha
Hello Akshitha
in else block
```

infinite while loop in python

```
In [34]: ▶ while(True):
    print("Akshitha")
    break
```

```
Akshitha
```

```
In [35]: ▶ for i in range(1,5):
    for j in range(i):
        print(i,end=" ")
    print()
```

```
1
2 2
3 3 3
4 4 4 4
```

```
In [36]: ▶ for i in range(1,10):
    if i==5:
        break
    print(i)
```

```
1
2
3
4
```

Loop control statements

break

stops the iteration

```
In [37]: ► for i in range(1,10):  
          if i==3:  
              break  
          print(i)
```

1
2

```
In [38]: ► for i in range(1,51):  
          if i%2==0:  
              continue  
          print(i)
```

1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49

```
In [39]: ► for i in range(1,11):  
          if i%5==0:  
              print(i)
```

5
10

```
In [40]: ▶ for i in range(1,10):  
          if i%5==0:  
              break  
          print(i)
```

```
1  
2  
3  
4
```

continue

skips the current iteration and move to the next iteration

```
In [41]: ▶ for i in range(1,10):  
          if i==3:  
              continue  
          print(i)
```

```
1  
2  
4  
5  
6  
7  
8  
9
```

```
In [42]: ▶ for letter in "geeksforgeeks":  
          if letter=="e" or letter=="s":  
              continue  
          print('current letter:',letter)
```

```
current letter: g  
current letter: k  
current letter: f  
current letter: o  
current letter: r  
current letter: g  
current letter: k
```

```
In [43]: ▶ for letter in "geeksforgeeks":  
          if letter=="e" or letter=="s":  
              break  
          print('current letter:', letter)
```

```
current letter: g
```

Pass

A null statement. It does nothing, just acts as a placeholder to avoid syntax errors.

```
In [44]: ▶ for i in range(1, 6):  
          if i == 3:  
              pass # does nothing, just a placeholder  
          print(i)
```

```
1  
2  
3  
4  
5
```

```
In [45]: ▶ available=5  
x=int(input('How many chocolates you want: ?'))  
i=1  
while i<=x:  
    if i>available:  
        break  
    print('chocolate')  
    i+=1
```

```
How many chocolates you want: ?6  
chocolate  
chocolate  
chocolate  
chocolate  
chocolate
```

```
In [46]: ▶ available=5  
x=int(input('How many chocolates you want: ?'))  
i=1  
while i<=x:  
    if i<available:  
        print('chocolate')  
    else:  
        print('no')  
    i+=1
```

```
How many chocolates you want: ?5  
chocolate  
chocolate  
chocolate  
chocolate  
no
```

```
In [47]: ▶ available=5
x=int(input('How many chocolates you want: ?'))
i=1
while i<=x:
    if i>available:
        print("out of stock")
        break
    print('chocolate')
    i+=1
```

How many chocolates you want: ?5
chocolate
chocolate
chocolate
chocolate
chocolate

```
In [48]: ▶ available=5
x=int(input('How many chocolates you want'))
i=1
while i<=x:
    if i>available:
        print("out of stock")
        break
    print('chocolate')
    i=i+1
print('bye for now')
```

How many chocolates you want5
chocolate
chocolate
chocolate
chocolate
chocolate
bye for now

```
In [49]: ▶ available=5
x=int(input('How many chocolates you want'))
i=1
while i<=x:
    if i>available:
        print("out of stock")
        break
    print('chocolate')
    i=i+1
print('bye for now')
```

How many chocolates you want4
chocolate
chocolate
chocolate
chocolate
bye for now

```
In [50]: ► for i in range(1,11):  
          print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

```
In [51]: ► for i in range(1,11):  
          if i==6:  
              break  
          print(i)
```

```
1  
2  
3  
4  
5
```

```
In [52]: ► for i in range(1,11):  
          if i==3:  
              continue  
          print(i)
```

```
1  
2  
4  
5  
6  
7  
8  
9  
10
```

```
In [53]: ► for i in range(1,11):  
          if i==6:  
              continue  
          print('hello:',i)
```

```
hello: 1  
hello: 2  
hello: 3  
hello: 4  
hello: 5  
hello: 7  
hello: 8  
hello: 9  
hello: 10
```



```
In [54]: ▶ for i in range(1,51):  
          if i%3==0:  
              print(i)  
          print('end')
```

```
3  
6  
9  
12  
15  
18  
21  
24  
27  
30  
33  
36  
39  
42  
45  
48  
end
```

```
In [55]: ► for i in range(1,51):  
           if i%3==0:  
               continue  
           print(i)  
           print('end')
```

```
1  
2  
4  
5  
7  
8  
10  
11  
13  
14  
16  
17  
19  
20  
22  
23  
25  
26  
28  
29  
31  
32  
34  
35  
37  
38  
40  
41  
43  
44  
46  
47  
49  
50  
end
```

you need to print the number from 1 to 50 but dont print the number which is divisible by 3 or 5

```
In [56]: ▶ for i in range(1,51):  
          if i%3==0 or i%5==0:  
              continue  
          print(i)
```

```
1  
2  
4  
7  
8  
11  
13  
14  
16  
17  
19  
22  
23  
26  
28  
29  
31  
32  
34  
37  
38  
41  
43  
44  
46  
47  
49
```

```
In [57]: ► for i in range(1,51):  
           if i%3==0 or i%5==0:  
               continue  
           print(i)  
           print('end')
```

```
1  
2  
4  
7  
8  
11  
13  
14  
16  
17  
19  
22  
23  
26  
28  
29  
31  
32  
34  
37  
38  
41  
43  
44  
46  
47  
49  
end
```

i dont want to print the values which are even numbers that means print only odd numbers

```
In [58]: ▶ for i in range(1,51):  
           if i%2==0:  
               continue  
           print(i)  
print('end')
```

```
1  
3  
5  
7  
9  
11  
13  
15  
17  
19  
21  
23  
25  
27  
29  
31  
33  
35  
37  
39  
41  
43  
45  
47  
49  
end
```

PRINTING PATTERN IN PYTHON

```
In [59]: ▶ print('# # # #')  
          print('# # # #')  
          print('# # # #')  
          print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [60]: ▶ for i in range(1,5):  
           print('#')
```

```
#  
#  
#  
#
```

```
In [61]:  ▶ for i in range(1,5):  
           print('#',end=" ")
```

```
# # # #
```

```
In [62]:  ▶ for i in range(1,5):  
           i=i+1  
  
           print('#')
```

```
#  
#  
#  
#
```

```
In [63]:  ▶ for i in range(1,5):  
           i=i+1  
           print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [64]:  ▶ for i in range(1,5):  
           if i<=5:  
               print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [65]:  ▶ for i in range(4):  
           print('#')
```

```
#  
#  
#  
#
```

```
In [66]:  ▶ for j in range(4):  
           print('#',end=" ")
```

```
# # # #
```

```
In [67]: ▶ for j in range(4):  
           print('#',end=" ")  
           for j in range(4):  
               print('#', end=" ")  
           print()
```

```
# # # # #  
# # # # #  
# # # # #  
# # # # #
```

```
In [68]: ▶ for j in range(4):  
           print('#', end=" ")  
  
           print()  
  
           for j in range(4):  
               print('#', end=" ")
```

```
# # # #  
# # # #
```

```
In [69]: ▶ for j in range(4):  
           print('#', end=" ")  
  
           print()  
  
           for j in range(4):  
               print('#', end=" ")  
  
           print()  
  
           for j in range(4):  
               print('#', end=" ")  
  
           print()  
  
           for j in range(4):  
               print('#', end=" ")
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [70]: ► for i in range(5):  
           for j in range(4):  
               print('#',end=" ")  
           print()
```

```
# # # #  
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [71]: ► for i in range(4):  
           for j in range(4):  
               print('#',end=" ")  
           print()
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [72]: ► for i in range(4):  
           for j in range(i+1):  
               print('#',end=" ")  
           i=i+1  
           print()
```

```
#  
# #  
# # #  
# # # #
```

```
In [73]: ► for i in range(1,5):  
           print('# ' * i)
```

```
#  
# #  
# # #  
# # # #
```

```
In [74]: ► for i in range(1,5):  
           for j in range(4):  
               if i>j:  
                   print("#",end=" ")  
           print()
```

```
#  
# #  
# # #  
# # # #
```



```
In [75]: ▶ for i in range(4):
           for j in range(i):
               print('#', end=" ")
           print()
```

```
#
# #
# # #
```

```
In [76]: ▶ for i in range(0):
           for j in range(i):
               print('#', end=" ")
           print()
```

```
In [77]: ▶ for i in range(4):
           for j in range(i+1):
               print('#', end=" ")
           print()
```

```
#
# #
# # #
# # # #
```

```
In [78]: ▶ for i in range(4):
           for j in range(4-i):
               print('#', end=" ")
           print()
```

```
# # # #
# # #
# #
#
```

```
In [79]: ▶ nums=[12,14,18,21,25,30,35]
           for i in nums:
               if i%2==0:
                   print(i)
```

```
12
14
18
30
```

```
In [80]: ▶ nums=[12,14,18,21,25,30,35]
           for i in nums:
               if i%5==0:
                   print(i)
```

```
25
30
35
```

In [81]: `nums = [12,14,18,21,25,20]`

```
for num in nums:
    if num % 5 == 0:
        print(num)
```

25
20

In [82]: `nums = [12,14,18,21,25,20,10]`

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
```

25

In [83]: `nums = [10,14,18,21,5,10]`

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break #it will
```

10

In [84]: `nums = [7,14,18,21,23,27]` *#hear there is no number which is divisible by 5 we*

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
```

In [85]: `nums = [7,14,18,21,23,27,29]` *#hear there is no number which is divisible by 5*

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
else:
    print('Number Not Found')
```

Number Not Found

```
In [86]: ► nums = [7,14] #hear there is no number which is divisible by 5 we got output
for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Number Not Found')
```

Number Not Found

Number Not Found

```
In [87]: ► nums = [7,14,18,21,23,27] #hear there is no number which is divisible by 5 we
for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Number Not Found')
```

Number Not Found

```
In [88]: ► nums = [10,14,18,21,20,27] #hear there is no number which is divisible by 5 w
for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Not Found')
```

10

```
In [89]: ► nums = [10,14,18,21,20,27,30] #hear there is no number which is divisible by
for num in nums:
    if num % 5 == 0:
        print(num)
        #break
    else:
        print('Not Found')
```

10

20

30

Not Found

```
In [90]: ▶ nums = [10,14,18,21,20,27] #hear there is no number which is divisible by 5 w
for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Not Found')
```

10

```
In [91]: ▶ num=14
for i in range(2,14):
    if i%2==0:
        print('not prime numbers')
    else:
        print('prime')
```

not prime numbers
prime
not prime numbers
prime
not prime numbers
prime
not prime numbers
prime
not prime numbers
prime
not prime numbers
prime

```
In [92]: ▶ num=14
for i in range(2,14):
    if num%i==0:
        print('not prime numbers')
    else:
        print('prime')
```

not prime numbers
prime
prime
prime
prime
not prime numbers
prime
prime
prime
prime
prime
prime

```
In [93]: ▶ num=14
for i in range(2,14):
    if num%i==0:
        print('not prime numbers')
        break
    else:
        print('prime')
```

not prime numbers

```
In [94]: ▶ num = 13

for i in range(2,num):
    if num % i == 0:
        print('Not prime Number')
    else:
        print('Prime Number')
```

Prime Number

Right angle triangle pattern

```
In [95]: ▶ for i in range(1,6):
           print('* ' * i)
```

```
*
* *
* * *
* * * *
* * * * *
```

```
In [96]: ▶ for i in range(1,6):
           for j in range(i):

               print('*',end=" ")
           print()
```

```
*
* *
* * *
* * * *
* * * * *
```

Inverted right angle triangle pattern

```
In [97]: ▶ for i in range(6):  
           for j in range(5-i):  
               print("*",end=" ")  
           print()
```

```
* * * * *  
* * * *  
* * *  
* *  
*
```

```
In [98]: ▶ for i in range(5,0,-1):  
           print('* ' * i)
```

```
* * * * *  
* * * *  
* * *  
* *  
*
```

```
In [1]: ▶ count=0  
         while(count<3):  
             count=count+1  
             print("Hello world")
```

```
Hello world  
Hello world  
Hello world
```

```
In [2]: ▶ age=28  
         while age>18:  
             break  
             print()
```

```
In [3]: ▶ age
```

```
Out[3]: 28
```

```
In [3]: ▶ i=0
a="geeksforgeeks"
while i<len(a):
    if a[i]=="e" or a[i]=="s":
        i+=1
        continue

    print(a[i])
    i+=1
```

g
k
f
o
r
g
k

```
In [5]: ▶ i=0
a="geeksforgeeks"
while i<len(a):
    if a[i]=="e" or a[i]=="s":
        i+=1
        break
    print(a[i])
    i+=1
```

g

```
In [6]: ▶ a="geeksforgeeks"
i=0
while i<len(a):
    i+=1
    pass
print("vallue of i:",i)
```

vallue of i: 13

```
In [7]: ▶ i=0
while i<4:
    i+=1
    print(i)
else:
    print("No break")
```

1
2
3
4
No break

```
In [8]: ▶ i=0
while i<4:

    print(i)
    i+=1
else:
    print("No break")
```

```
0
1
2
3
No break
```

```
In [9]: ▶ i=0
while i<4:

    print(i)
    i+=1
    break
else:
    print("No break")
```

```
0
```

```
In [10]: ▶ i=0
while i<4:
    i+=1
    print(i)
    break
else:
    print("No break")
```

```
1
```

```
In [12]: ▶ s=["Geeks", "FOr", "geeks"]
for i in s:
    print(i)
```

```
Geeks
FOr
geeks
```

```
In [13]: ▶ s="geeks"
for i in s:
    print(s)
```

```
geeks
geeks
geeks
geeks
geeks
```



```
In [14]: ▶ for i in range(0,10,2):  
          print(i)
```

```
0  
2  
4  
6  
8
```

```
In [15]: ▶ # control statements with for loop
```

```
for i in "environment":  
    if i=="i" or i=="m":  
        continue  
    print(i)
```

```
e  
n  
v  
i  
r  
o  
n  
m  
e  
n  
t
```

```
In [16]: ▶ for i in "environment":  
          if i=="i" or i=="m":  
              break  
          print(i)
```

```
e  
n  
v
```

```
In [17]: ▶ for i in "environment":  
          pass  
          print(i)
```

```
t
```

```
In [18]: ▶ for i in range(1,4):  
          print(i)  
        else:  
          print("NO break")
```

```
1  
2  
3  
NO break
```

When you loop through a list or tuple you only get values but you also want index use `enumerate()` function

```
In [19]: ► li=["eat","sleep","repeat"]
          for i,j in enumerate(li):
              print(i,j)
```

```
0 eat
1 sleep
2 repeat
```

```
In [20]: ► li=["eat","sleep","repeat"]
          for i,j in enumerate(li):
              print(i)
```

```
0
1
2
```

```
In [21]: ► for i in range(1,4):
          for j in range(1,4):
              print(i,j)
```

```
1 1
1 2
1 3
2 1
2 2
2 3
3 1
3 2
3 3
```

```
In [26]: ► a=["shirts",'sock','pants','sock','towel']
          b=[]
          for i in a:
              if i=="sock":
                  continue
              else:
                  print("washing",i)
          b.append("socks")
          print(f"washing {b}")
```

```
washing shirts
washing pants
washing towel
washing ['socks']
```

```
In [27]: ▶ for i in range(1,8):  
          d=3+(i-1)*0.5  
          print(f"Day {i}: Run {d:.1f} miles")
```

```
Day 1: Run 3.0 miles  
Day 2: Run 3.5 miles  
Day 3: Run 4.0 miles  
Day 4: Run 4.5 miles  
Day 5: Run 5.0 miles  
Day 6: Run 5.5 miles  
Day 7: Run 6.0 miles
```

```
In [28]: ▶ x=[1,2]  
          y=[4,5]  
          for i in x:  
              for j in y:  
                  print(i,j)
```

```
1 4  
1 5  
2 4  
2 5
```

```
In [29]: ▶ x=[1,2]  
          y=[4,5]  
          i=0  
          while i<len(x):  
              j=0  
              while j<len(y):  
                  print(x[i],y[j])  
                  j=j+1  
              i=i+1
```

```
1 4  
1 5  
2 4  
2 5
```

```
In [32]: ▶ i=2
for i in range(2,3):
    for j in range(1,11):
        print(i,"*",j,"=",i*j)
    print()
```

```
2 * 1 = 2
2 * 2 = 4
2 * 3 = 6
2 * 4 = 8
2 * 5 = 10
2 * 6 = 12
2 * 7 = 14
2 * 8 = 16
2 * 9 = 18
2 * 10 = 20
```

```
In [36]: ▶ N=5
for i in range(5,6):
    for j in range(1,11):
        print(i,"*",j,"=",i*j)
```

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

```
In [41]: ▶ N=5
for j in range(1,11):
    print(N,"*",j,"=",N*j)
```

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

```
In [42]: N=5
for j in range(1,11):
    print(N,"*",j,"=",N*j)
```

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

```
In [65]: s="DoctorPhenomenal"
for i in range(0,15,2):
    print(s[i],end="")
```

```
DcoPeoea
```

```
In [67]: s="geeks"
for i in range(0,5,2):
    print(s[i],end="")
```

```
ges
```

```
In [22]: x=3
for i in range(x):
    for j in range(x-i):
        print(i,end=" ")
    print()
```

```
0 0 0
1 1
2
```

```
In [1]: x=3
while x>=0:
    print(x)
    x=x-1
```

```
3
2
1
0
```

```
In [3]: ▶ x=5
while x>=0:
    print(x)
    x=x-1
```

```
5
4
3
2
1
0
```

```
In [3]: ▶ x=10
i=1
while(i<=10):
    square=i**2
    if i>3:
        break
    print(square,end=" ")
    i=i+1
```

```
1 4 9
```

```
In [5]: ▶ n=9
for i in range(0,11):
    i=i+1
    print(n,"*",i,"=",n*i)
```

```
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
9 * 11 = 99
```

```
In [3]: ▶ sum=0
n=3

for i in range(1,n+1):
    sum=sum+i
print(sum)
```

```
6
```

```
In [7]: ▶ sum_loop = 0
        for i in range(1, n + 1):
            sum_loop += i
        print("Sum using loop:", sum_loop)
```

Sum using loop: 6

```
In [3]: ▶ x=0
        sum=0
        n=3
        while x<=n:
            sum=sum+x
            x=x+1
        print(sum)
```

6

```
In [18]: ▶ n1=9
          n2=4

          for i in range(1,11):
              mul_1=n1*i
              mul_2=n2*i
              print(mul_1-mul_2,end=" ")
```

5 10 15 20 25 30 35 40 45 50